

Websites are now offering more dynamic content and new features with unbelievable performance for website users. The features and rich content have now become the sole concern for website developers. Several internet studies about website performance have been conducted in the past years. They suggest that dynamic features and faster loading results attract more users, engagement, and sales. In a nutshell, as a developer, you must know how to build a faster website that can be accessed by users in a few seconds.

Area of knowledge

Quicker website responses and performance come under the category of web performance optimisation. With the advancement of technology, numerous techniques, technologies, and practices have emerged to enhance the user's web experience. However, the main focus of these web performing technologies and practices is to mitigate the web page's download size, set up a limit for HTTP requests and, optimise JavaScript.

This chapter discusses the vast category of web performances which include speed of server response during a user request, the function of load testing, the steps which generate server response rate, and software in website load testing.

What is load testing?

In simple terms, website load testing determines the performance of an application or site under an absolute pre-calculated load. It means how a website or an app will behave while multiple net users access it simultaneously. The main aim of load testing is to ensure the safe, secure, and smooth functioning of software or site under real-time load conditions. However, load testing comes under non-functional testing. Moreover, it is a small part of performance testing.

What does load testing identify?

- It can identify the overall operating capacity of a website or an application.
- Load testing can figure out whether the on-going website structure is sufficient to maintain its smooth running.
- The sustainability of an app or a website during the maximum user access.
- How many users an application or a website can support
- A website's capacity for allowing more access.

As mentioned above, it is a subset of performance testing and comes under the non-functional testing category only. Load testing practices and methods may vary for web-based applications, server/client, and both the Internet and Intranet depending on the number of users or traffic.